

CFD 2025-26 – Lab 13

Exercise 1

Let us consider the Navier-Stokes problem defined in the domain Ω provided as `square-cyl.msh`:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{2}{\text{Re}} \varepsilon(\mathbf{u}) + \nabla p = \mathbf{0}, & \text{in } \Omega, \ t \in (0, T], \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, \ t \in (0, T] \end{cases}$$

where the Reynolds number Re is equal to 20 and the boundary conditions are given by:

- no-slip conditions on the boundary of the square obstacle;
- free-slip conditions on the lateral boundary of the pipe;
- a parabolic velocity profile on the inlet boundary Γ_{in} ;
- homogeneous Neumann boundary conditions on the outlet boundary Γ_{out} ,

and $\mathbf{u}_{\text{Stokes}}$ is the solution of the corresponding Stokes problem.

We want to discretize the problem using the stable pair of spaces $\mathbb{P}_b^1/\mathbb{P}^1$ and solve the problem using the algebraic Chorin-Temam projection scheme, with a semi-implicit treatment of the advection term.

- 1 Introduce the time-discrete formulation of the problem by the backward Euler scheme with a semi-implicit treatment of the advection term, and derive the algebraic formulation of the FEM applied to a time step of the scheme.

Notation: in the following, we are going to call C^{n+1} the matrix

$$C^{n+1} = \frac{1}{\Delta t} M + A + N(\mathbf{U}^n).$$

- 2 Modify the algebraic problem in order to use the incremental pressure $\delta p^{n+1} = p^{n+1} - p^n$ instead of the pressure p^{n+1} as the system variable. The dofs vector $\begin{bmatrix} \mathbf{U}^{n+1} \\ \delta \mathbf{P}^{n+1} \end{bmatrix}$ will be the solution of the modified algebraic problem.
- 3 Introduce the inexact factorization of the system matrix that corresponds to the algebraic, incremental Chorin-Temam projection scheme.
- 4 Consider (do not implement yet!) the resulting numerical scheme as an inexact factorization algebraic scheme, which is described by the following steps:

1) Solve the predictor step:

$$C^{n+1} \tilde{\mathbf{U}}^{n+1} = \mathbf{f}^{n+1} + \frac{1}{\Delta t} M \mathbf{U}^n - B^T \mathbf{P}^n.$$

2) Solve the Poisson problem for the pressure increment:

$$B M^{-1} B^T \delta \mathbf{P}^{n+1} = \frac{1}{\Delta t} B \tilde{\mathbf{U}}^{n+1}.$$

3) Projection step:

$$\mathbf{U}^{n+1} = \tilde{\mathbf{U}}^{n+1} - \Delta t M^{-1} B^T \delta \mathbf{P}^{n+1}.$$

4) Pressure update:

$$\mathbf{P}^{n+1} = \mathbf{P}^n + \delta \mathbf{P}^{n+1}.$$

Notice that in step 2, M^{-1} is considered as an approximation of the velocity-velocity block matrix in the Schur complement.

Re-write the algorithms using an *incremental approach* also on velocity and implement it exploiting the `fieldsplit` preconditioning features of PETSc.